

Repayment Norms and Risk Rationing in Microfinance

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Introduction

Motivation

- Conventional lending: physical collateral → mitigate information problems.
 - Quantity rationing: many poor households lack the assets that lenders require, and are denied credit access. (Stiglitz et al., 1981)
 - Risk rationing: collateral-eligible individuals voluntarily limit borrowing to avoid the risk of losing their physical or financial assets. (Boucher et al., 2008)
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Table: Credit Rationing in Conventional and Microfinance Lending.

Country	Rationing Type (%)			Physical Collateral
	Quantity	Risk	Price	
Peru	37	19	46	Yes
Honduras	23	16	62	Yes
Nicaragua	48	12	40	Yes
Bangladesh	26	20	54	No

Sources: Boucher et al., 2008; Exploratory Survey 2024.

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⇒ Use of non-physical collateral may have some characteristics as physical collateral, leading to a similar level of incidence of quantity and risk rationing.

Why Does Default Risk Matter Without Physical Collateral?

- Early *Joint Liability MF* induced peer monitoring and assortative matching
(Besley et al., 1995; Ghatak, 2001; Stiglitz, 1990)
- Modern *Individual Liability MF* continues using enforcement mechanisms, similar to informal lending, that leverage existing community ties and interpersonal relationships
(Feigenberg et al., 2013; Rai et al., 2010)
- Dynamic incentives: future credit eligibility depends on repayment history (Giné et al., 2010)
- **Internal norms:** borrowers may intrinsically value honoring contractual obligations
- **Reputational norms:** credit transactions happen in socially observable settings
- I focus on the role of norms in individual liability microfinance, independent of future credit incentives.



This Study

Broad Research Question

How important are norms in reducing default in individual liability microfinance?

Specific objectives:

- Measure the relative contribution of internal and reputational norms.
- Examine whether repayment norms explain risk rationing behavior in microfinance (ongoing analysis)

Mapping Norms to Experiment

Mapping Norms to Experimental Design

I measure repayment response when only internal and reputational norms are made salient in a framed lab-in-the-field experiment.

Norm ^a	Microfinance Context	Experimental Setup
<i>Intrinsic Motivation or Personal Norm:</i> Belief about what one ought to do, based on an internalized self-standard	<i>Internal Norm:</i> Borrower's internalized self-standard to honor contractual obligation	<i>Private Game (P):</i> Borrowers privately decide how much to repay, reminded only of the debt obligation.
<i>Reputational Motivation or Social Norm:</i> Belief about perceived social expectations that concerns social approval or disapproval	<i>Reputational Norm:</i> Borrower's motivation to maintain reputation based on perception of social expectation	<i>Revelation Game (R):</i> Borrowers are informed that their default will be revealed to the peers.

^a Definitions draw on Bénabou et al., 2006, Akerlof, 1980, Cialdini et al., 1991, and Andreoni et al., 2009.

Model: Repayment with Internal and Reputational Norms

- Borrower chooses repayment $\rho \in [0, \bar{R}]$ after observing income $y(\theta)$

$$U(\rho) = \underbrace{u(y(\theta) - \rho)}_{\text{monetary payoff}} + \underbrace{\alpha_i N(\rho - \rho^I(\theta))}_{\text{penalty for violating internal norm}} + \underbrace{s \cdot \gamma_i N(\rho - \rho^R(X_i))}_{\text{penalty for violating reputational norm}}$$

- $y(\theta) - \rho$: retained income after repayment
- $u(\cdot)$: increasing, concave function
- $\rho^I(\theta)$: internalized belief of how much she should repay in a given state
- $\alpha_i \geq 0$: valuation of acting in accordance with the internal standard
- $\rho^R(X_i)$: perceived belief of what others expect of her to repay
- $\gamma_i \geq 0$: sensitivity to maintaining social standing
- $s \in \{0, 1\}$: whether default is socially observable
- $N(z) = N(\min\{z, 0\})$: deviation from perceived norm generates penalty

Empirical mapping:

Private Game ($s = 0$): Internal norm response $\rightarrow$ measured as repayment without externally induced penalty

Revelation Game ($s = 1$): Reputational norm response $\rightarrow$ identified through variation in social observability

Experimental Setup

Experimental Setup

- Sample: 919 microfinance borrowers engaged in agriculture from 100 villages
- Rice farming activity
 - Loan size: 30,000
 - Total debt obligation: 35,000
 - Yield depends on state of nature
 - Borrowers record repayment decisions for the private and revelation games
 - Remaining money after repayment in the games is converted into actual earnings, from random draws



Experimental Setup: Decision Sheets

  Income  Principal + Interest 				
Decision	Repay	Default	Remaining Money	Actual Earning
	35,000 	0	45,000 	450
	30,000 	5,000 	50,000 	500
	25,000 	10,000 	55,000 	550
	20,000 	15,000 	60,000 	600
	15,000 	20,000 	65,000 	650
	10,000 	25,000 	70,000 	700
	5,000 	30,000 	75,000 	750
	0	35,000 	80,000 	800

Experimental Setup: Revelation Game

Name of the Borrower	Amount Defaulted
Farhana Akter	
Shyamoli Rani Das	
Mousumi Akter	
Nipa Begum	
Sharmin Sultana	
Monira Mondol	
Tania Begum	
Nahida Parvin	
Rukaiya Haque	
Sanjida Siddika	
Purnima Sarker	

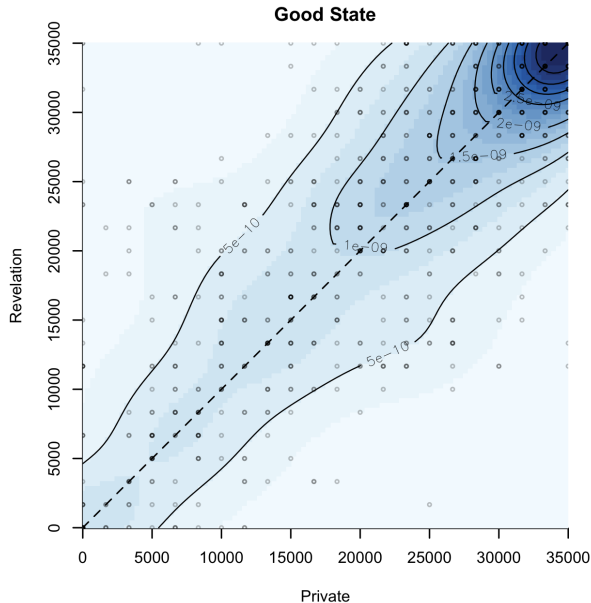
Experimental Design

	Practice		Actual Game						Payoff Determination and Revelation
	Round 1		Round 2		Round 3		Round 4		
	G1	G2	G3	G4	G5	G6	G7	G8	
Sequence 1	P	R	R	P	P	R	R	P	Draw round and state of nature
Sequence 2	R	P	P	R	R	P	P	R	

- Each session follows the same structure.
- Each game consists of three states of nature: Normal, Good, and Bad yield
- Strategy method: borrowers choose repayment amount for all games, in all states
- We do not reveal the state of nature of the individual to the peers.

Empirical Strategy

Joint Distribution of Private and Revelation Game Repayments





Empirical Specification: OLS with Fixed Effects

Within subject identification:

$$y_{igr} = \beta \mathbf{1}\{g = R\}_{igr} + \mu_i + \lambda_r + \varepsilon_{igr} \quad (1)$$

- The indicator $\mathbf{1}\{g = R\}_{igr}$ equals one for the revelation game
- β identifies effect on repayment due to social observability
- y_{igr} : Amount repaid by individual i in game g in round r ;
- μ_i : Individual fixed effect;
- λ_r : Round fixed effect;

Results: OLS with Fixed Effects

Table: Effect of Social Observability on Loan Repayment.

	(1) <i>Good Yield Repayment</i>	(2) <i>Normal Yield Repayment</i>	(3) <i>Bad Yield Repayment</i>	(4) <i>Expected Repayment</i>
Default Revelation	767.14*** (211.20)	797.97*** (221.73)	562.21*** (198.31)	717.99*** (145.82)
Constant	23,108.45*** (105.60)	22,629.67*** (110.86)	17,261.52*** (99.15)	21,162.86*** (72.91)
Observations	5,514	5,514	5,514	5,514
R-squared	0.69	0.66	0.73	0.80
Game Rounds	2–4	2–4	2–4	2–4
Fixed Effect	Individual, Round	Individual, Round	Individual, Round	Individual, Round
SE Cluster	Individual	Individual	Individual	Individual

Standard errors clustered at the individual level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Outcome Distribution

Empirical Specification: Two-limit Tobit

- Repayment is censored at bounds $[0, \bar{Y}]$
- **Latent repayment model:**

$$y_{igr}^* = \beta^T \mathbf{1}\{g = R\}_{igr} + \mathbf{x}_{igr}' \boldsymbol{\psi} + \varepsilon_{igr},$$

$$y_{igr} = \begin{cases} 0 & \text{if } y_{igr}^* \leq 0 \\ y_{igr}^* & \text{if } 0 < y_{igr}^* < \bar{Y} \\ \bar{Y} & \text{if } y_{igr}^* \geq \bar{Y} \end{cases}$$

- β^T identifies the effect of social observability on the latent repayment
- $\mathbf{x}_{igr}$ is a vector of experimental controls

Results: Two-limit Tobit

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Panel A: Coefficients				
Default Revelation	1,604.96*** (436.77)	1,247.97*** (407.09)	793.23** (316.11)	862.53*** (180.54)
Constant	30,080.11*** (891.29)	27,447.11*** (774.71)	19,079.55*** (692.14)	22,715.92*** (455.51)
Error Variance	587,869,972.93*** (35,422,927.48)	450,239,667.50*** (24,847,824.90)	391,423,291.47*** (21,140,334.70)	175,454,563.47*** (8,468,293.88)
Observations	5,514	5,514	5,514	5,514
Uncensored	2,608	2,958	3,383	4,331
Left-censored	626	581	932	281
Right-censored	2,280	1,975	1,199	902
Fixed Effect	Game Seq., Round	Game Seq., Round	Game Seq., Round	Game Seq., Round
SE Cluster	Individual	Individual	Individual	Individual

Standard errors clustered at the individual level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Censoring limits are $[0, 35,000]$ BDT in all columns. $\hat{\sigma}^2$ is the estimated variance of the latent error term.

All specifications control for game sequence order and round.

Results: Two-limit Tobit Marginal Effects

	(1) <i>Good Yield Repayment</i>	(2) <i>Normal Yield Repayment</i>	(3) <i>Bad Yield Repayment</i>	(4) <i>Expected Repayment</i>
Panel B: Marginal Effects				
$\frac{\partial}{\partial x} E[y^* x]$ (latent, unconstrained)	1,605*** (436.7)	1,247*** (82.14)	793.8** (196.4)	862.6*** (141.0)
$\frac{\partial}{\partial x} E[y x]$ (observed, censored)	768.2*** (207.3)	678.3*** (407.1)	494.1** (316.1)	675.5*** (80.21)
$\frac{\partial}{\partial x} E[y^* x, 0 < y^* < \bar{Y}]$ (latent, interior)	254.4*** (68.61)	252.0*** (220.8)	186.2** (74.16)	383.8*** (180.5)
Observations	5,514	5,514	5,514	5,514
Uncensored	2,608	2,958	3,383	4,331
Left-censored	626	581	932	281
Right-censored	2,280	1,975	1,199	902
Controls	Game Seq., Round	Game Seq., Round	Game Seq., Round	Game Seq., Round
SE Cluster	Individual	Individual	Individual	Individual

- **Latent AME:** the latent repayment response absent censoring
- **Observed AME:** effect on observed repayment, measures what changes in practice
- **Interior AME:** effect among borrowers whose latent repayment falls strictly between the bounds

Norm Heterogeneity and Risk Rationing

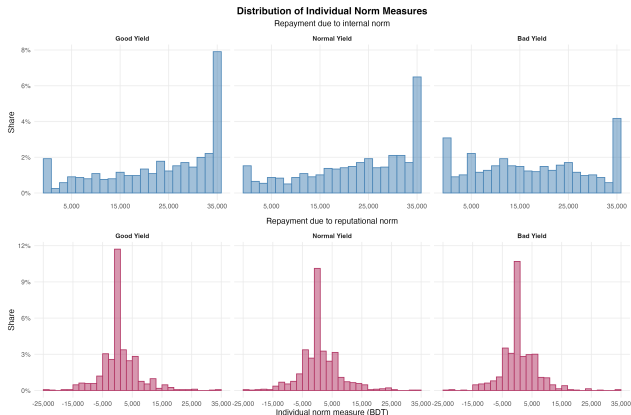
Norm Heterogeneity and Risk Rationing

- The results establish that social observability increases repayment, larger once censoring is accounted for.
- But the average effect masks heterogeneity: Outcome Distribution
 - some borrowers repay more even when default is private;
 - some respond strongly when default is socially observable;
 - some are non-responsive to observability.
- To connect the experimental results to the risk rationing outcomes, I next recover individual-specific measures of internal and reputational repayment norms.

Recovering Individual Norm Measures

A fixed-effects specification allows the revelation response to vary across borrowers, where μ_i reflects internal norm response and β_i reflects reputational norm response.

$$y_{igr} = \mu_i + \beta_i \mathbf{1}\{g = R\}_{igr} + \mathbf{x}'\psi + \varepsilon_{igr},$$

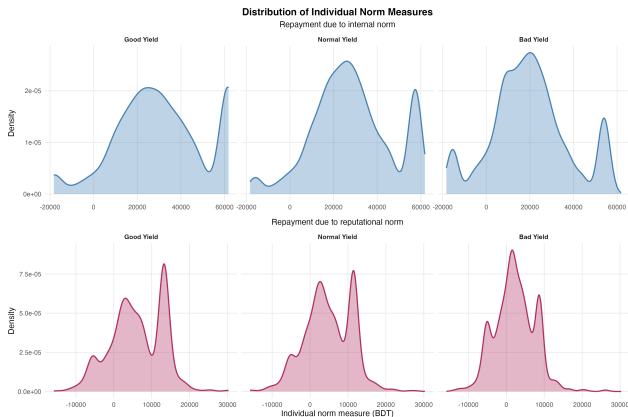


Recovering Individual Norm Measures (Latent Outcome)

Latent-variable specification accounts for repayment censoring, where μ_i^* reflects latent internal norm response and β_i^* reflects latent reputational norm response.

$$y_{ir}^{P*} = \mu_i^* + \mathbf{x}'\psi + \varepsilon_{ir}^P, \quad y_{ir}^P = \max\{0, \min\{\bar{Y}, y_{ir}^{P*}\}\},$$

$$\tilde{y}_{ir}^* = \beta_i^* + \mathbf{x}'\psi + \varepsilon_{ir}, \quad \tilde{y}_{ir} = y_{ir}^R - \bar{y}_i^P = \max\{-\bar{y}_i^P, \min\{\bar{Y} - \bar{y}_i^P, \tilde{y}_{ir}^*\}\}.$$



Norms and Risk Rationing

- Microfinance shows risk rationing rates comparable to conventional lending, although secured without physical collateral.
- Following Boucher et al., 2008, risk-rationed borrowers may self-ration credit due to default risk, even when the expected return to borrowing is positive.
- Heterogeneity in norm-driven repayment responses suggests that borrowers may differ in their perceived default risk.
- In ongoing analysis, I refine the individual-specific norm measures accounting for censoring and link these to household survey data.
- I examine whether borrowers with stronger internal or reputational repayment norms are more likely to be risk rationed.

Concluding Remarks

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- **Reputational norms matter for repayment.** Social observability raises latent repayment by 800–1,600 taka depending on income state, larger than OLS once censoring is accounted for, and independent of dynamic credit access incentives.
- **Substantial heterogeneity in norm sensitivity.** Both internal and reputational norm responses vary widely across borrowers and individual-specific norm measures may predict risk rationing in microfinance.
- **Broader implication: norms as non-physical collateral.** If normative enforcement plays an analogous role to physical collateral, borrowers with strong repayment norms may face more default risk. Introducing risk management tools could improve credit allocation efficiency.

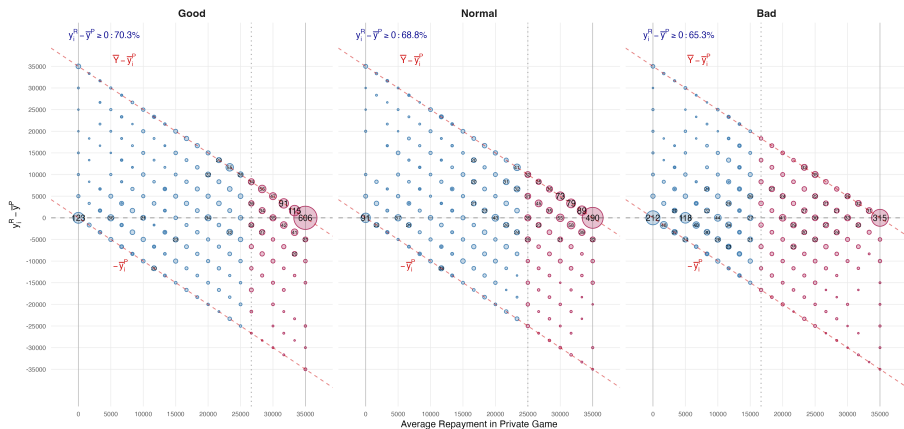


Thank you!

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Multilevel Censoring Problem



- Both private and revelation games repayment are bounded by the feasible choice set $[0, 35,000]$.
- Borrowers whose private repayment lies on either bound have limited scope to respond to revelation condition.

Empirical Specification: Variable-limit Tobit

- Borrowers near $[0, \bar{Y}]$ cannot fully express reputational responses
- Revelation Game outcomes are censored relative to individual baseline behavior
- **Relative repayment:** $\bar{y}_i^P = \frac{1}{n_i^P} \sum y_{ir}^P$, $\tilde{y}_{ir} = y_{ir}^R - \bar{y}_i^P$
- **Individual-specific bounds:** $-\bar{y}_i^P \leq \tilde{y}_{ir} \leq \bar{Y} - \bar{y}_i^P$
- **Tobit with varying limits:**

$$\tilde{y}_{ir}^* = \beta^{VT} + \mathbf{x}_{ir}' \boldsymbol{\psi} + \varepsilon_{ir},$$

$$\tilde{y}_{ir} = \begin{cases} -\bar{y}_i^P & \text{if } \tilde{y}_{ir}^* \leq -\bar{y}_i^P \\ \tilde{y}_{ir}^* & \text{if } -\bar{y}_i^P < \tilde{y}_{ir}^* < \bar{Y} - \bar{y}_i^P \\ \bar{Y} - \bar{y}_i^P & \text{if } \tilde{y}_{ir}^* \geq \bar{Y} - \bar{y}_i^P \end{cases}$$

β^{VT} : the latent reputational effect accounting for individual-specific censoring.